

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination
November

CL 205 Building Materials and Construction

Date: 30.11.2013,

Time: 01:30 a.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) i. What will be number of steps if the floor height is 3.0 m and riser height is 20 cm? 03
- ii. Write maximum number of steps in a flight.
- iii. What is the name of stair in which provided steps are of wedge shape winders?
- (b) Sketch quarter turn staircase with nominal dimensions and labels. 04
- Q - 2 (a) Explain Flemish bond with plan and elevation for a 1 brick thick wall. 05
- (b) Sketch ashlar facing and brick backing type of composite masonry. 04
- (c) Write materials used for plasterwork. What are the requirements of good plaster? 05
- OR
- Q - 2 (a) i. Which brick masonry bond gives more attracting and pleasing appearance? English or Flemish 02
- ii. Which bond provides more strength for 2 brick thick wall?
- (b) Sketch various brick bats used in brick masonry work. 03
- (c) Write classification of stone masonry. Sketch random rubble stone masonry. 04
- (d) Explain stucco plaster work. 05
- Q - 3 (a) Sketch double leaf four paneled door with elevation, horizontal and vertical section details and labeled components. 05
- (b) Sketch the type of window which is provided for the sole purpose of light and vision only. 04
- (c) Sketch roof truss with label for span of 8.0 m. 05

OR

- Q - 3 (a) Sketch the type of door which is suitable for a bank building. 05
- (b) Sketch the window which provides extra space in the room and improves the appearance. 04
- (c) Sketch roof truss with label for span of 12.0 m. 05

SECTION - II

- Q - 4 (a) A brick wall 30 cm thick having height 3.2 m above plinth level. Plinth level of structure is at 0.45 m from ground level having same thickness of wall. Wall carries superimpose load of 100 kN/m. Unit weight of soil 16.5 kN/m^3 , Angle of repose 30° and q_s is 158 kN/m^2 . Foundation have cement concrete base, which has unit weight of 23.5 kN/m^3 , modulus of rupture is equal 245 kN/m^2 . Consider unit weight of masonry 19.5 kN/m^3 . 07

Using given data calculate following parameters for strip footing of wall.

- a) Pressure at base of footing
- b) Width of footing
- c) Depth of footing as per Rankines formula
- d) Depth of footing as per uniform angle of sprayed from masonry.

- Q - 5 (a) State classification of bricks. What are the chemical changes occur during burning of brick? 06
- (b) Name various types of glass. What are the attribute of glass for using it in glazing. 06
- (c) Offset of cement concrete block, which is under masonry sprayed footing, is 15cm, Bearing pressure on soil is 98.75 kN/m^2 and modulus of rupture is 245 kN/m^2 . Calculate depth of concrete block. 02

OR

- Q - 5 (a) What are the ingredients of concrete? List the desirable properties of concrete. 06
- (b) What are the selection criteria of wood as building material? State various defects in wood. 06
- (c) State the causes of failure of foundation. 02

- Q - 6 (a) Explain physical properties of Engineering Material. 05
- (b) List types of floors. Draw neat sketch of two way R.C.C. slab as a floor. 05
- (c) What is underpinning? State purpose of under pinning. 04

OR

- Q - 6 (a) Explain mechanical properties of Engineering Material 05
- (b) Draw neat Figure of Stone slab floor with steel joist. 05
- (c) Write the names only for component parts of scaffolding. Draw neat sketch of mason's scaffolding. 04